MIL-B-52239C
12 February 1985
SUPERSEDING
MIL-B-52239B(ME)
29 April 1976

MILITARY SPECIFICATION

BAR, CONNECTING, BRIDGE: STEEL:

FOR PONTON FLOAT, HALF SECTIONAL,

PNEUMATIC, 12-TON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a steel connecting bar to be used with a 12-ton, pneumatic, ponton float.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, for a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-R-52243	- Retainer, Bridge Pin.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS A2.4 - Symbols for Welding and Nondestructive Testing Including Brazing.
ANSI/AWS A5.1 - Covered Carbon Steel Arc Welding Electrodes.
ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 519 - Seamless Carbon and Alloy Steel Mechanical Tubing.
ASTM A 575 - Merchant Quality Hot-Rolled Carbon Steel Bars.
ASTM B 633 - Electrodeposited Coatings of Zinc on Iron and Steel.
ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distrubuted among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The connecting bar shall be as shown on figure 1 and as specified herein. The connecting bar shall be fabricated from one continuous length of tubing, with a steel flange at one end and a hemispherically shaped nosepiece at the other end, both of which shall be welded to the connecting bar.

3.2 First article. When specified (see 6.2), a sample connecting bar shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The connecting bar shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

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3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.1.3 Tubing. The connecting bar shall be round seamless tubing conforming to ASTM A 519, grade 1015 to 1025, hot finished, annealed.

3.3.1.4 Bar steel. The connecting bar flange and hemispherical nosepiece shall be steel conforming to ASTM A 575, grade M1015 to M1025.

3.3.1.5 Welding electrodes. Welding electrodes shall conform to ANSI/AWS A5.1, E 7014 to E 7028.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the connecting bar may be newly fabricated from recovered materials to the maximum extent practicable, provided the connecting bar produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the connecting bar.

3.4 Retainer. The connecting bar shall be provided with a retainer conforming to MIL-R-52243, length 5-9/16 inches.

3.5 Treatment. The completed connecting bar shall be cleaned, treated, and zinc coated in accordance with ASTM B 633, type II, Fe/Zn13.

3.6 Metal stamping. Each connecting bar shall be stamped as shown in figure 1, and as specified (see 6.2). Metal stamping shall be performed on the connecting bars prior to application of zinc coating and shall be legible after the coating has been applied.

3.7 Workmanship.

3.7.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners.

3.7.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the connecting bar to conform to the examination requirements specified in section 4. Parts to be joined by

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fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch. Parts joined by groove welds shall be ground smooth and finished to the diameter of the parent metal. Unless otherwise specified (see 6.2), the welding process used in fabrication of the connecting bars shall be at the option of the contractor.

3.7.3 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

Certification shall be made available for inspection by the contracting officer or designated representitive, upon request.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article connecting bar inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article connecting bar inspection.

4.3.1 First article samples. One or more connecting bars (see 6.2) shall be taken from the first ten connecting bars produced.

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4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The connecting bar shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the connecting bar.
106. Dimensions not as specified.
107. Components missing or not as specified.
108. Zinc coating missing or not as specified.
109. Identification marking missing, incomplete, or not legible.
110. Workmanship not as specified.
111. Welding certification not as specified.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQl shall be 2.5 percent defective.

112.	Boxes not as specified.
113.	Strapping not as specified.
114.	Commercial packing not as specified.
115.	Marking missing, incomplete, or not legible.

5. PACKAGING

5.1 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A. The connecting bars with retainer pins shall be packed in quantities of 12 in close-fitting boxes conforming to PPP-B-621, class 2, or PPP-B-601, overseas type, grade B. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.1.2 Level B. The connecting bars with retainer pins shall be packed in quantities of 12 in close-fitting boxes conforming to PPP-B-621, class 1, or PPP-B-601, domestic type, grade B. Strapping of the boxes shall be as specified in the box specification appendix.

5.1.3 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.2 Marking.

5.2.1 Levels A and B. Levels A and B marking shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. In addition, the gross weight and cube shall be marked on each container.

6. NOTES

6.1 Intended use. The connecting bar is intended to be used to fasten together two pneumatic ponton floats and to connect the outrigger beam to the ponton float.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of connecting bars required (see 3.2, 4.3, and 6.3).

c. Additional data required for metal stamping (see 3.6).

d. When the welding process will not be at the option of the contractor (see 3.7.2).

e. Degree of packing required (see 5.1).

f. When finish A strapping is required (see 5.1.1).

6.3 First article. When a first article inspection is required the item shall be examined and should be a first produced connecting bar. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

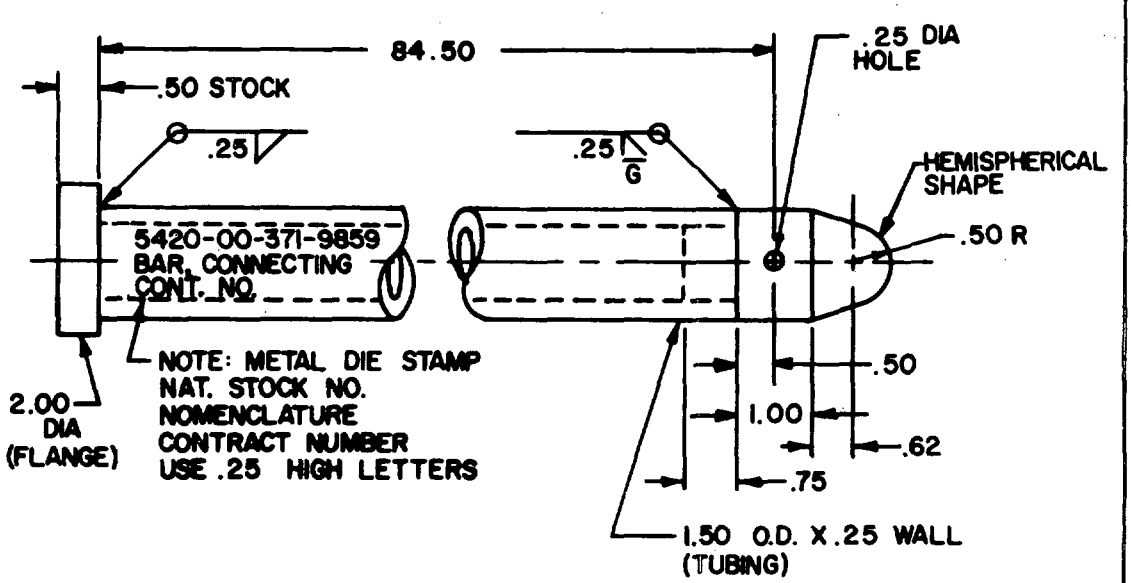
Custodian:
Army - ME

Review activity:
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Preparing activity:
Army - ME

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES
2. UNSPECIFIED TOLERANCE: ±.03
3. WELDING SYMBOLS ARE IN ACCORDANCE WITH ANSI/AWS A 2.4

FIGURE 1. Bar, connecting, bridge.

X2000B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-52239C	Bar, Connecting, Bridge: Steel: For Ponton Float, Half Sectional, Pneumatic, 12-Ton

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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